Rehabilitation Center Experience

Kennedy Krieger Institute | Neurobehavioral Unit (Inpatient)

585.617 Rehabilitation Engineering | Module 10

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Project Topic

under one minute

- **Wheelchair-mounted interface for handheld powered yard tools** — string trimmers, edgers, hedge trimmers, blowers.
- **Population:** manual wheelchair users with full upper-limb function, maintaining their own residential yard on unimproved ground.
- **Scope:** an interface and mounting problem. Not a new tool, not a new wheelchair.
- **The governing difficulty is that three constraints are coupled:**
  - The chair is the only available reaction mass for tool torque, and it has a limited stability envelope.
  - The working envelope must reach ground level and hedge height from a fixed seated anthropometry.
  - Manual propulsion is itself two-handed, so holding a tool occupies the limbs mobility requires.
- No design has been selected. The visit was for input on the problem, not approval of a solution.

The Center

Kennedy Krieger Institute, Baltimore, Maryland

- Internationally recognized for pediatric **brain, spinal cord and musculoskeletal disorders**; treats individuals **up to age 21**.
- The inpatient rehabilitation unit is a **25-bed unit**, accredited by **CARF** and the **Joint Commission**.
- Patients receive **several hours a day** of occupational, physical and speech therapy.
- **I visited the Neurobehavioral Unit (Inpatient)**, a specialist service for children and adolescents with intellectual and developmental disabilities and **severe problem behavior** — self-injury, aggression, property destruction.
  - Treatment is applied behavior analysis: functional analysis to identify what maintains a behavior, then function-based treatment.
  - Admissions are measured in weeks to months, not days.

Patient Observation

what I was able to see

- I observed a **behavioral treatment session** on the unit rather than a physical or occupational therapy session.
- What stood out was not the clinical content but **how the session was structured**:
  - Behavior was recorded continuously and in real time, not summarized afterward.
  - Staffing was high, and a second staff member was present purely for safety.
  - The environment itself was part of the treatment — surfaces, fixtures and equipment all chosen for what they prevent.
- **Patient privacy limited what I could observe directly**, and I did not record or photograph anything on the unit.

What the Observation Changed

impact on my preliminary design

- **Measurement is treated as part of the intervention.** Nothing was judged by impression. My PDS should carry **measurable acceptance criteria**, not qualitative aims — e.g. task completion time, number of attempts, setup time.
- **Staff presence is a design variable.** A task needing a second person is a different product from one that does not. For my project the equivalent question is whether the user can deploy and stow the tool **without a second pair of hands**.
- **Safety is designed into the environment before it is trained into the person.** My instinct had been to treat stability as something the user manages by technique. The better framing is to design so the unsafe state is hard to reach.

Technology Overview

what is used, and how often

- **Used constantly, deliberately low-tech:** visual schedules, token boards, timers, picture cards, protective equipment, padded surfaces.
- **Used constantly, sophisticated:** real-time behavioral data collection on tablets, and video monitoring of sessions.
- **Used selectively, sophisticated:** augmentative and alternative communication devices, including speech-generating devices.
- **The most sophisticated element was methodological, not physical** — the functional analysis protocol itself. The hardware supporting it is simple on purpose.
- **Durability is the dominant hardware requirement.** Equipment here must survive deliberate destruction, not just wear.

Relevance to My Project

nothing directly related, and why that matters

- **I saw no technology related to my project** — no wheelchair-mounted tooling, no outdoor task equipment, and few manual wheelchair users.
- **That absence is structural, not an oversight.** The NBU treats behavior; it is not a mobility service, and its population is not mine.
- **But the constraints transfer, and they are harder than a rehab gym would impose:**
  - **Abuse tolerance** — designed against deliberate destruction, not incidental wear.
  - **Cleanability** — a requirement, not a preference.
  - **No ligature points, no entrapment, nothing readily detached or thrown.**
  - **Engagement as an engineering requirement** — if the user disengages, the device has failed, whatever its mechanics.
- If my interface can meet that bar, residential yard use is a far easier environment.

Care Provider Interviewed

Daniel Dobin, Clinical Assistant II

- **Clinical Assistant II, Neurobehavioral Unit (Inpatient)**, Kennedy Krieger Institute — eighteen months on the unit, now **training incoming clinical assistants** alongside direct patient care.
- **Johns Hopkins University undergraduate**, 2024–2028.
- **Research Fellow, Johns Hopkins Center for Suicide Prevention**; first author on systematic reviews published in the *Journal of the American Academy of Child and Adolescent Psychiatry* and *Neurotherapeutics* (2026).
- **Crisis Text Line crisis counselor**, two years.
- **Why this background was useful:** he sees equipment in daily use, he trains the people who use it, and he reads evidence critically — so he distinguished what is claimed about a device from what is observed.

Interview Themes (1 of 2)

constraints and priorities

- **Safety requirements come before function.** Anything brought onto the unit is screened for ligature risk, pinch points, parts that detach, and anything that can be thrown. These are pass/fail, not trade-offs.
- **Cleanability sits with safety**, not with convenience.
- **Asked to rank needs, the ordering was roughly:**
  - **Essential** — safety under limited supervision; durability; cleanability.
  - **Important** — speed of setup; whether it holds the user's attention.
  - **Lower** — cost, which matters but is rarely what kills a device.
- **Cost ranking lowest was the surprise.** My assumption had been that affordability was the binding constraint.

Interview Themes (2 of 2)

failure and solution directions

- **How devices actually fail:** breakage under deliberate force, small parts lost, setups too slow to be worth starting, and patients simply disengaging. **Mechanical failure was not the most common mode.**
- **Success depends on the staff member as much as the device.** Some equipment only works when a particular person is on shift — a dependency that never appears in a specification.
- **On my three solution directions**, the preference was for **reducing the force at source** over carrying it on the chair, on the grounds that fewer attachments means fewer failure points and faster setup.
- **I asked for a referral** to a physical therapist, occupational therapist or seating and mobility specialist, since stability margins, propulsion and funding sit outside this role.

Impact on My Preliminary Design

what changes as a result

1. Quick-release moves from nice-to-have to essential. Safety-driven, and it was the clearest single message of the visit.

2. Abuse tolerance and cleanability enter the PDS as requirements, not as finishing details.

3. Abandonment becomes a design criterion. If the setup is slow enough that the user stops bothering, the device has failed. Setup time gets a target.

4. Acceptance criteria become measurable — attempts, elapsed time, setup time — following the unit's own practice of measuring rather than judging.

5. Scope confirmed, source corrected. The problem framing holds, but the quantitative inputs — stability margin, propulsion penalty, funding route — must come from an adult mobility clinician, not this unit.

What I Could Not Gather, and Next Steps

- **No client interview.** The NBU is a pediatric inpatient behavioral unit; patient privacy and the clinical setting made a client conversation inappropriate. The user perspective is therefore still missing.
- **No quantitative mobility data.** Tipping margins, propulsion penalties and reimbursement pathways are outside a direct-care behavioral role.
- **Next steps:**
  - Follow up with a PT, OT or seating and mobility specialist for the quantitative inputs.
  - Interview adult manual wheelchair users directly — the population my project actually targets.
  - Carry the needs ranking into a Kano diagram for the Preliminary Design Review.
  - Use the safety and durability constraints as screening criteria in concept generation.